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Master's Thesis in Informatics: Games Engineering

Something with Statecharts and PIX:E

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Irgendwas mit Statecharts und PIX:E

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Abstract

PaceMaker is a web-based tool that assists video game developers to plan and realize player experience in a video game. Visualization is based on statecharts, however, some functionality described in statecharts is not yet realized, as the use case for some aspects of statechart's specification remain uncertain. In this thesis we aim to research what aspects of statecharts make sense to implement and then implement them in PIX:E, the successor of PaceMaker. Furthermore, we want to investigate whether statecharts are the most suitable concept to model player experience and therefore research other concepts. We compare these concepts based on objective values to make an informed decision about what concept PIX:E should use going forward.

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1 Introduction

The aim of this thesis proposal is to give an overview of the thesis' goals. It provides a baseline for supervisor, advisor, the student, and anyone else interested in this thesis for future discussions.

1.1 Motivation

Video games, just like other media, aim to create an experience for the player [1]. One of the major differences between video games and other media, like film, television, or music, comes from the player's ability to directly interact with the video game. As a consequence of this interaction, the player experience is non-linear. As video game developers can only design the rules of the game, the direct influence of the game developer on the player's experience is more limited compared to other media. Salen and Zimmermann describe this as a "second-order design problem" [2]. This also allows for a more obvious distinction between the artifact, and the experience of the consumer for video games compared to other non-interactive forms of media [1].

Still, the video games industry uses terminology and principles seen in other entertainment industries. The usage of concepts from other disciplines arises from game development being an emergent discipline [2]. Furthermore, video game development itself is an interdisciplinary undertaking, which can require experts in programming, story writing, art, psychology, and more [1]. Common terminology coming from other disciplines when designing a game include beats from music [3] and story-structures, such as the three-act-structure or the hero's journey, from screenwriting [4, 5]. However, the aforementioned interactivity provided by video games requires novel approaches to plan the player's experience to then realize it in the game [6].

1.2 PaceMaker

PaceMaker is a web application that gives the user tools to support video game development around pacing as a concept more deliberately. Currently, the tool supports three key tasks:

- specifying experience with multiple properties,

- assigning the experience specifications to beats and using them to create experience charts, and
- showing an automatically generated pacing diagram based on nodes selected in the experience chart using Dijkstra's algorithm.

The tool uses the Vue3 framework. The reasons for developing PaceMaker as a web-application were platform- and engine dependency. PaceMaker's development aimed at implementing functional requirements, while usability concerns have not yet been carefully addressed. However, PaceMaker does not realize the full specification of statecharts as described in the initial paper by Harel [7]. Instead, the current implementation resembles the specification of state diagrams, not statecharts, which is an extension of state diagrams. [3]

2 Problem Statement

Video game development needs to account for interactivity and the player's experience when playing the game. Bond and Beale [8] analyzed 33 game reviews and found that variety was among the four most important features for a game to be considered good. Their sampling criteria for variety includes "non linearity, choice, differences, [...]"[8, p.420], among others. We draw that the interactivity is not just any property of a game, but a key factor of why people even play games. Video game development does not just have to deal with interactivity, but see it as an opportunity to enhance the player's experience. However, interactivity adds a complex factor to the planning stage of video game development and tools need to accommodate interactivity. This claim is further supported as design and tools ranked as the most and fifth-most common problem during development in an analysis of 200 postmortems, respectively [9].

PaceMaker aims to solve the issue of planning for interactivity to allow game developers to plan their games more thoroughly by providing a tool. While PaceMaker allows to model player experience with state diagrams, scalability issues arise as hierarchy (state-levels) is not yet implemented. This makes planning longer games with PaceMaker less attractive for game developers, limiting its potential usage. Additionally, we can not yet make an informed decision on whether orthogonality has a valid use case in video game development or whether there are other vital functionalities missing by adhering to the statechart specification.

During PaceMaker's development, using a simple form of statecharts was the result of the functional requirements. However, there might be other concepts than statecharts suitable for representing player experience. Finding and comparing more concepts that are already used to measure plan experience of entertainment to statecharts can yield valuable results going forward.

To summarize, the following questions need to be addressed in the thesis:

- What applications do state-levels and orthogonality have in planning player experience during game development?
- Given that there are other suitable approaches to modeling player experience, how do they compare in objective measurements (e.g., runtime for comparable actions)?

3 Approach

As mentioned, PaceMaker aims to help game developers to introduce pacing into the current workflow. As of writing this proposal, the developers of PaceMaker, now called PIX:E, work on a new version of the tool using a new framework—Nuxt. It builds on the previously used framework Vue3 and offers additional tools for developers, like automatic routing between pages and more plugins, among others. The functionality of the old version needs to be ported.

3.1 Literature Review

The first objective in our literature review is to find out what kind of concepts other researchers and professionals use to plan consumer or customer experience. We do not restrict our approach to video game development, as initial searches restricted to video game development on Google Scholar did not yield many useful results. Therefore, we will conduct an integrative review as described by Snyder [10] to find common themes among different fields that can be used to plan experience. This part of the thesis aims to inform the reader about experience planning in general, not tied to video game development, and allows us to make more informed decisions about what features PIX:E requires.

Second, we want to investigate whether there are other approaches to model player experience than statecharts. For this, we will conduct a systematic literature review in order to find most approaches that could be an alternative to statecharts, also based on our findings in the first part of the literature review. For this objective, we will also conduct research with non-scientific articles, specifically blogs about video game development processes. We think that the game development industry provides valuable input that scholars should take into account, just like scholars can provide valuable input for the video game development industry, and aim to bridge the gap between academics and practitioners [11].

3.2 Development of PIX:E

The development of PIX:E is dependent on multiple factors. First, specific development goals of PIX:E may change over the course of working on this thesis. While the bigger objective—introducing functionality to work with modeling player experience in PIX:E—remains, the way to do that might change. Still, the first few weeks of development will focus on porting code from Vue3 to Nuxt, so that the base functionality of PaceMaker is also in PIX:E. Frequent meetings with the other developers on the project ensure being on time, writing clean code, and allow changes during development. Implementing new features, as a result of recommendations from the literature review, will be agreed on with the advisor.

As to the validation of the implementation with statecharts, we aim to implement other concepts that allow for modeling player experience. By comparing these approaches with multiple use cases (no experience chart, a basic experience chart, an experience chart of a simple level, a complex experience chart) regarding their runtime, we will determine whether statecharts remain a good solution to our problem.

3.3 Limitations

When it comes to the limitations of this thesis, we need to address the following points. The initial paper on PaceMaker refers to addressing usability as future work, however, the aim of this thesis is not to test PIX:E's usability. We are also aware that the initial research for other alternatives to model player experience might not yield any alternatives. In this case, we will either compare different implementations of statecharts, or we have to adjust the goal of the thesis altogether.

Comparing runtimes between different algorithms and concepts can be skewed if the implementation quality of the algorithms is not the same, for instance [12]. Therefore, while it is a goal of the thesis, we can only make informed conclusions if the implementations of the algorithms are optimized for the use case, which seems difficult, given the time frame of the thesis and the other goals. Therefore, the results of this comparison might only be first tendencies, but need more work in another thesis.

4 Literature Review

As mentioned in section 3.1, the thesis will not only draw information from scientific works. This literature review chapter reflects this by not only referring to scientific works, but also solutions to modeling player experience in the video game development industry. The structure of this literature review is not optimal, but this chapter acts mainly as a basis for further research for the thesis.

Ohwada et al. [13] propose an advanced technique for route planning based on Dijkstra's algorithm, which also finds use in PaceMaker. While they take into account factors that are not important for video game development, such as fast pass options, the advanced technique of wayfinding in a graph might be beneficial for generating pacing diagrams in PIX:E.

The development of Cyberpunk 2077 [14] utilizes a custom-written scene and quest system [15]. Especially of interest to the goal the thesis is the ability to have graphs inside graphs (i.e., nestedness/state-levels), clear overviews for large graphs, and tools that allow to work on the graphs, such as timelines [15]. As we want to develop a game development tool that game developers use, findings in studies investigating the requirements of such a tool can provide a good basis for research, like the one by Kasurinen et al. [16].

In a first search, we found approaches that model game behavior, but not necessarily player experience, using WorkFlow nets (based on Petri nets) [17] or finite state machines with an extension for hierarchical relationships [18]. Further investigating these concepts, whether they can be used for modeling player experience, and researching differences between them and statecharts can be used as a base to add more nuance to the comparison between these concepts.

The initial specification of statecharts [7] has several problems, with [19] stating them and comparing multiple variants of statecharts that aim to solve these problems. The list of problems can be used to make statements about whether statecharts are suitable for modeling player experience or not.

Other scientific works on player experience focus on how to predict player experience without humans [20], adjusting narrative based on the player's previous input [21], or the connection between audio and player experience [22].

5 Schedule

As seen in chapter 3, there are two main areas of work: developing PIX:E and research + writing. As I am going to work on those in parallel, this section contains two timetables to reflect the concurrency of the thesis.

5.1 Development of PIX:E

Time Period	Task Description	Artifact
19.05.2025- 22.06.2025	porting from Vue3 to Nuxt framework	PIX:E with the same functionality as PaceMaker
23.06.2025- 10.08.2025	implementing functional requirements based on research	PIX:E with implemented functional requirements
10.08.2025- 28.09.2025	implementing other concepts in PIX:E	reasonably efficient and fast implementation of other concepts for measurements
29.09.2025- 12.10.2025	evaluation of implemented concepts	measurement data for each concept and action
13.10.2025- 10.11.2025	buffer time for implementation; nice-to-have features	buffer time

5.2 Research and Writing

Time Period	Task Description	Artifact
19.05.2025-07.06.2025	exploratory Research—concepts for modeling player experience, other entertainment industry	list of concepts with pros and cons
08.06.2025-22.06.2025	researching use cases for orthogonality and state-levels in modeling player experience	list of use cases and implementation proposals
23.06.2025-06.07.2025	thesis writing about use cases for orthogonality and state-levels	finished chapter on use cases for orthogonality and state-levels with reasoning for implementation details
07.07.2025-27.07.2025	thesis writing about exploratory research	finished chapter on findings and recommendations of exploratory research for PIX:E development
18.08.2025-07.09.2025	thesis writing about implementation of statechart functionality	finished chapter functionality
15.09.2025-12.10.2025	description of other concepts and their implementations	finished chapter
13.10.2025-26.10.2025	thesis writing on evaluation	finished chapter with recommendations based on the findings
27.10.2025-16.11.2025	finish thesis writing; iterating and feedback gathering	finished master's thesis
17.11.2025-19.11.2025	printing thesis & enjoying Paulaner	submission

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